

Lakshya JEE 2.0 2025

Maths

Determinants

DPP: 3

Q1

If $\Delta_r = \begin{vmatrix} 2r-1 & {}^m C_r & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2 m & \sin m^2 \end{vmatrix}$, then

the value of $\sum_{r=0}^m \Delta_r$ is

- (A) 1 (B) 3
(C) 2 (D) 0

Q2

Let $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$ where p is a

constant. Then $\frac{d^3}{dx^3} \{f(x)\}$ at $x = 0$

- (A) p
(B) $p + p^2$
(C) $p + p^3$
(D) Independent of p

Q3

If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix}$, then the ordered

$= (A+Bx)(x-A)^2$

pair (A, B) is equal to

- (A) $(4, 5)$ (B) $(-4, -5)$
(C) $(-4, 3)$ (D) $(-4, 5)$

Q4 For the determinant

$\Delta = \begin{vmatrix} -1 & a & a^2 \\ \sin x & \cos y & \sin x \\ \cos y & \sin x & \cos y \end{vmatrix} \quad \left(a \in \mathbb{R} \right)$

- (A) If Δ is non-zero, it is independent of a
(B) Δ is always independent of a
(C) If Δ is independent of a , then $\sin x + \sin y$ can be equal to 3

(D) None of these

Q5

If $\Delta_1 = \begin{vmatrix} 10 & 4 & 3 \\ 17 & 7 & 4 \\ 4 & -5 & 7 \end{vmatrix}$, Δ_2

$= \begin{vmatrix} 4 & x+5 & 3 \\ 7 & x+12 & 4 \\ -5 & x-1 & 7 \end{vmatrix}$

such that $\Delta_1 + \Delta_2 = 0$, then

- (A) $x = 5$
(B) x has no real value
(C) $x = 0$
(D) $x = 1$

Q6

The value of $\sum_{n=1}^N U_n$ if

$U_n = \begin{vmatrix} n & 1 & 5 \\ n^2 & 2N+1 & 2N+1 \\ n^3 & 3N^2 & 3N \end{vmatrix}$ is

- (A) 0 (B) 1
(C) -1 (D) 2

Q7

If $\Delta = \begin{vmatrix} 1 & 3 \cos \theta & 1 \\ \sin \theta & 1 & 3 \cos \theta \\ 1 & \sin \theta & 1 \end{vmatrix}$, then the

$\frac{1}{1000} [\text{maximum value of } \Delta - \text{minimum value of } \Delta]^3$ is equal to ____.

- (A) 1 (B) -1
(C) 0 (D) 2

Q8


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Let
$$\begin{vmatrix} x^2 + 3x & x - 1 & x + 3 \\ x + 1 & -2x & x - 4 \\ x - 3 & x + 4 & 3x \end{vmatrix} = ax^4 + bx^3 + cx^2 + dx + e$$

be an identity in x , where a, b, c, d, e are independent of x . Then the value of e is

- (A) 4 (B) 0
(C) 1 (D) none of these

Q9 Let $px^4 + qx^3 + rx^2 + sx + t$

$$= \begin{vmatrix} x^2 + 3x & x - 1 & x + 3 \\ x + 1 & -2x & x - 4 \\ x - 3 & x + 4 & 3x \end{vmatrix}$$

be an identity, where p, q, r, s and t are constants, then the value of s is equal to

- (A) 70 (B) 71
(C) 72 (D) 73

Q10 If
$$\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$$
 then

- (A) a, b, c are in A.P.
(B) a, b, c are in G.P.
(C) α is a root of $ax^2 + 2bx + c = 0$
(D) $(x - \alpha)$ is a factor of $ax^2 + 2bx + c = 0$

Q11 If
$$\Delta_r = \begin{vmatrix} 2^{r-1} & 2 \cdot 3^{r-1} & 4 \cdot 5^{r-1} \\ x & y & z \\ 2^n - 1 & 3^n - 1 & 5^n - 1 \end{vmatrix},$$
 then

$\sum_{r=1}^n \Delta_r$ is equal to

- (A) xyz (B) 1
(C) -1 (D) 0

Q12 If
$$\Delta_r = \begin{vmatrix} r & x & \frac{n(n+1)}{2} \\ 2r - 1 & y & n^2 \\ 3r - 2 & z & \frac{n(3n-1)}{2} \end{vmatrix},$$
 then

$\sum_{r=1}^n \Delta_r$ is equal to

- (A) xyz
(B) $nxyz$

- (C) 0
(D) none of these

Q13 If
$$\Delta_r = \begin{vmatrix} 1 & r & 2^r \\ 2 & n & n^2 \\ n & \frac{n(n+1)}{2} & 2^{n+1} \end{vmatrix},$$
 then the value of

$\sum_{r=1}^n \Delta_r$ is

- (A) n
(B) $2n$
(C) $-2n$
(D) n^2

Q14 If
$$I_n = \begin{vmatrix} 1 & k & k \\ 2n & k^2 + k + 1 & k^2 + k \\ 2n - 1 & k^2 & k^2 + k + 1 \end{vmatrix}$$

and $\sum_{n=1}^k I_n = 72$, then $k =$

- (A) 8 (B) 9
(C) 6 (D) none of these

Q15 If
$$D_k = \begin{vmatrix} 1 & n & n \\ 2k & n^2 + n + 2 & n^2 + n \\ 2k - 1 & n^2 & n^2 + n + 2 \end{vmatrix}$$

and $\sum_{k=1}^n D_k = 48$, then n equals

- (A) 4
(B) 6
(C) 8
(D) None of these



Answer Key

Q1 (D)

Q2 (D)

Q3 (D)

Q4 (D)

Q5 (A)

Q6 (A)

Q7 (A)

Q8 (B)

Q9 (B)

Q10 (B, C, D)

Q11 (D)

Q12 (C)

Q13 (C)

Q14 (A)

Q15 (A)



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